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Optimizing internet of things security through blockchain enabled software defined networking

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The Internet of Things (IoT) faces increasing security challenges, as traditional encryption-based methods often fail to prevent cyberattacks while maintaining reliable and cost-effective communication. To overcome these issues, this work introduces a security model that enhances existing frameworks by identifying and analyzing malicious data flows, significantly improving IoT network protection. The proposed approach connects blockchain-based Software-Defined Network (SDN) controllers with SDN switches and IoT applications, creating a highly secure and energy-efficient communication system. This framework detects potential threats, minimizes security risks, and improves overall network performance. To support real-time monitoring, two types of messages are introduced: non-contemporary beacons, which inform SDN controllers about network events through blockchain-SDN switches, and encoded statistical beacons, which provide detailed insights for further security analysis. The proposed method demonstrates substantial improvements over existing technologies, including a 76% reduction in network overhead, 82% lower latency, 73% stronger resistance to malicious data flows, 54% higher throughput, 32% less bandwidth usage, 65% faster algorithm execution, and 41% lower memory consumption. These results highlight the effectiveness of the model in providing faster, safer, and more efficient IoT communication.

Blockchain was originally developed to protect data exchanges in distributed ledger systems. However, integrating it with the centralized structure of IoT networks has been challenging. Traditional IoT setups rely on cloud-based systems to control data flow across devices. While this approach supports basic communication, it struggles with scalability due to the rapid growth of IoT devices. Additionally, centralized architectures create security vulnerabilities, making it necessary to explore more secure and adaptable solutions[1].

By combining blockchain with IoT, a more secure and reliable framework can be created. This integration helps protect sensitive data, supports automated agreements, and improves network efficiency. Instead of relying on a single control point, IoT devices communicate through encrypted, decentralized channels. This method enhances security while reducing the chances of unauthorized access. Among various blockchain-based solutions, the Information for Operational and Tactical Analysis (IOTA) ledger has emerged as a promising standard for IoT transactions. It enables structured data exchanges and has gained recognition from major industry players[2].

In this context, SDN offers a smarter way to manage IoT infrastructure. SDN separates network control functions from physical devices, allowing centralized decision-making for better resource management. An

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SDN controller acts as the central hub, gathering network-wide information and applying predefined rules to direct traffic efficiently. Through direct communication with switches and routers, the controller optimizes traffic flow, balances network load, and quickly adapts to changing conditions[3]. This approach not only improves performance but also strengthens security by enabling advanced threat detection and access control mechanisms.

Overall, integrating blockchain with SDN creates a scalable and secure IoT framework. This combination enhances data privacy, reduces operational risks, and improves network performance, making IoT systems more efficient and resilient in handling modern security challenges.

Problem statement

The growing number of IoT devices has led to serious security challenges[4,5], as traditional encryption-based methods are not enough to prevent cyberattacks while maintaining reliable and cost-effective communication. Many existing security solutions rely on centralized models, making them vulnerable to large-scale attacks and single points of failure. Additionally, high computational costs and network inefficiencies limit their effectiveness in real-time IoT environments. Without an advanced security framework that can detect and counter malicious activities efficiently, IoT networks remain exposed to risks such as unauthorized access, data manipulation, and service disruptions.

Background and motivation

As IoT technology continues to expand into critical sectors like healthcare, smart cities, and industrial automation, security becomes a major concern. Traditional security frameworks struggle with real-time threat detection and scalability, leading to increased vulnerabilities. Blockchain and SDN provide promising solutions, but their full potential in IoT security has not been fully explored. By combining blockchain-based SDN controllers with IoT applications, a more secure, scalable, and energy-efficient communication system can be created. This work aims to enhance IoT security by developing a model that detects malicious data flows and improves overall network performance.

The paper's main objectives are as follows:

- Developed an advanced security model that detects and mitigates malicious data flows, enhancing IoT network protection.
- Designed a security framework that integrates blockchain-based SDN controllers with SDN switches and IoT applications for improved threat detection.
- Introduced a real-time monitoring mechanism using non-contemporary beacons and encoded statistical beacons to provide detailed security insights.
- Achieved significant performance improvements, including a 76% reduction in network overhead, 82% lower latency, 73% stronger resistance against malicious data flows, 54% higher throughput, 32% lower bandwidth usage, 65% faster algorithm execution, and 41% reduced memory consumption.
- Provided a scalable and energy-efficient security solution that enhances the reliability and cost-effectiveness of IoT communication.

The following section outlines the study's methodology structure: Section 2 discusses peer-competing existing protocols and their limitations. Section 3 presents the proposed methodology. The result analysis of both proposed and existing schemes is presented in Section 4. Section 5 concludes the study and explores possible future scope.

Literature review

The evolution of IoT, blockchain, and AI-driven technologies has led to various research efforts focused on security, privacy, resource management, and efficiency in smart systems. This section reviews key studies that have introduced innovative solutions for challenges in IoT-based environments, including smart cities, healthcare, edge computing, and vehicular networks. While these studies contribute to improving security, scalability, and performance, they also present certain limitations that require further exploration.

Recent research by[1] focuses on security and privacy challenges in smart city applications, proposing potential solutions to address major threats and security gaps in smart city networks. However, the study lacks implementation details and practical validation.[2] introduces the Tangle, a distributed ledger designed for IoT applications, improving scalability and reducing transaction costs compared to traditional blockchain solutions. Nevertheless, it still faces challenges in maintaining consistency and dealing with network congestion.[3] provides a comprehensive survey on mobile edge computing (MEC), discussing architecture, applications, and future directions. The work effectively categorizes MEC approaches but does not provide experimental validation for the proposed frameworks. To enhance energy-efficient routing in IoT,[4] presents a fuzzy ranking-based scheduling model that optimizes resource utilization and network lifespan, but its effectiveness in large-scale deployments remains unexplored.

To improve IoT network security and efficiency,[4] explores the integration of SDN and blockchain for IoT networks, offering valuable insights into security enhancements without proposing a concrete implementation model. Addressing security in online voting,[5] introduces a blockchain and SDN-IoT-based voting system, ensuring data integrity and transparency but requiring optimization for high transaction loads.[6] proposes a blockchain-based security framework for IoT-based SDN networks, effectively reducing unauthorized access and enhancing trust but at the cost of increased computational overhead. For secure access control in SDN-IoT networks,[7] presents SILedger, a blockchain and attribute-based encryption (ABE) mechanism, improving security but suffering from high computational complexity, which limits real-time performance.

Blockchain integration for IoT spectrum sharing is studied in[8], which introduces a dynamic spectrum allocation model to enhance efficiency. However, the model lacks performance evaluation in real-world scenarios. To secure financial transactions in IoT applications,[9] proposes a blockchain-enabled protocol ensuring transaction security and data integrity, but further optimization is required to reduce processing delays. Optimizing urban traffic management in IoT-based smart cities,[10] presents an intelligent traffic optimization model using SDN, improving network efficiency while lacking real-world testing in large-scale deployments.[11] introduces an SDN-controlled analytics framework for healthcare IoT systems, optimizing resource allocation and decision-making but failing to address security vulnerabilities in healthcare applications.

To improve concurrency control in fog-cloud environments,[12] proposes a transaction mechanism that enhances processing efficiency but suffers from synchronization issues under heavy workloads. A blockchain-enabled task offloading model for edge computing is introduced by[13], offering enhanced security but facing blockchain-related latency concerns.[14] develops a security-by-contract model for industrial IoT edge networks, ensuring contract-based security enforcement, though scalability remains a challenge. A broad survey on IoT architecture, protocols, and challenges is provided by[15], categorizing key aspects but failing to propose specific solutions to address the identified challenges.

AI-enhanced cloud-based IoT solutions are explored in[16], improving IoT efficiency but requiring high computational resources. To counter DDoS attacks in SDN-IoT,[17] presents an early detection mechanism using blockchain, significantly enhancing security at the cost of increased processing overhead.[18] introduces a distributed security framework for IoT using SDN and blockchain, offering strong attack resistance but facing deployment challenges. Addressing security in IoV applications,[19] proposes a blockchain-enabled IoV model for fog computing and 5G networks, ensuring improved security but suffering from scalability issues.

To enhance cybersecurity in SDN-IoT environments,[20] presents a blockchain-based security framework, improving protection against cyber threats while facing latency issues in real-time applications.[21] introduces a federated learning-based vehicle route optimization protocol for IoV, promoting sustainable transportation and efficient route planning. However, a robust communication model is needed to effectively handle federated learning updates in dynamic environments.

Research gaps identified in literature

Despite significant advancements in IoT security and privacy, several critical research gaps remain unaddressed. Existing privacy-preserving architectures, such as those designed for smart city models[7], lack comprehensive analysis regarding scalability and real-time deployment in dynamic environments. While data protection mechanisms like wavelet transform-based partitioned histogram approaches[8] improve accuracy, they do not adequately address computational overhead and adaptability in large-scale IoT systems. Similarly, biometric-based authentication mechanisms[9] enhance security in medical IoT devices, yet they fail to consider vulnerabilities related to biometric data storage and processing. Trust management models for Fog-based IoT (FIoT)[11] effectively evaluate user trustworthiness but lack robust mechanisms to handle dynamic trust fluctuations in large-scale deployments. Additionally, while message authentication mechanisms[12] improve privacy, they require further validation regarding computational efficiency and feasibility for lightweight IoT devices. Privacy-preserving techniques for mobile IoT services[13] demonstrate improved security but have not been thoroughly analyzed for the trade-offs between accuracy and processing delay.

Moreover, existing security models for cyber-attack mitigation[11,15] primarily focus on smart homes and healthcare but do not adequately address challenges in large-scale, heterogeneous IoT networks. Although homomorphic encryption mechanisms[16] enhance data privacy, their high computational complexity makes real-time processing impractical for resource-constrained IoT devices. While attribute-based encryption mechanisms[17] provide security, they lack dynamic access control solutions for multi-user IoT environments. Location privacy solutions[19] improve security; however, their applicability to real-time tracking and navigation-based IoT applications remains uncertain. Blockchain-based lightweight architectures[2,5] effectively reduce storage costs, but the trade-off between storage efficiency and transaction processing speed requires further evaluation. Furthermore, scalable blockchain structures[7] optimize transaction processing, yet they lack energy-efficient designs suitable for constrained IoT environments.

Authentication schemes[8] enhance security but fail to address interoperability issues across heterogeneous IoT networks. Similarly, threshold-based IoT security services[14,16] exhibit reduced response time but require further assessment in large-scale networks with dynamic user behavior. Finally, while attribute-based access control models[18,19] provide enhanced security, they do not offer comprehensive protection against evolving attack strategies in IoT environments. These gaps highlight the need for further research in designing scalable, computationally efficient, and adaptive security mechanisms for heterogeneous IoT networks.

Table 1 shows a comparison between past research and the proposed model. It lists the methods, main ideas, common issues, and what the new model improves. Many existing studies introduce smart security features using blockchain or SDN. However, these often have issues like high system load, slow performance, or no real testing. The new model works better by lowering delay, saving energy, using less memory, and handling attacks in real time. This comparison shows what the model adds to help current IoT systems run faster and safer.

Proposed method

The proposed methodology integrates blockchain nodes within the SDN framework, allowing direct interaction with SDN controllers to strengthen security and authentication in network communication. These blockchain nodes validate and store transactions, creating a tamper-resistant environment where all network traffic managed by SDN controllers is authenticated before processing. To facilitate communication within this framework, two key protocols are utilized: the southbound protocol, which connects SDN controllers to network devices such as switches and routers, and the northbound protocol, which handles interactions between blockchain nodes and

References	Method Used	Main Focus	Common Issues	Improvements in Proposed Model
[1]	Security review for smart cities	Highlights security problems and gives ideas for solutions	No working model or real test	Uses a tested security setup for IoT
[2]	Tangle-based ledger for IoT	Cuts costs and supports more users	Faces slow performance and data issues	Uses an SDN-blockchain setup for better results
[3]	Study on Mobile Edge Computing	Lists edge computing types and future plans	No experiments or test results	Adds real implementation with focus on security
[4]	Fuzzy method for IoT routing	Saves resources and boosts lifespan	Not tested in large networks	Uses routing that adjusts in real-time
[5]	Blockchain and SDN for e-voting	Keeps data safe in online voting	Takes too long to process many votes	Speeds up transactions with less load
[6]	Blockchain security for SDN-IoT	Blocks unwanted access and builds trust	Needs a lot of computing power	Uses lightweight security to reduce system load
[7]	Blockchain and ABE for access control	Strengthens access security	Not good for real-time tasks	Adds faster access control that uses fewer resources
[8]	Blockchain for IoT frequency sharing	Improves use of spectrum in networks	No real-world tests done	Adds a working model that handles spectrum well
[9]	Blockchain in financial IoT	Keeps transaction data safe	Processing takes too long	Speeds up secure processing
[10]	Smart traffic using SDN	Improves traffic flow in cities	Not tested in large cases	Works well with high traffic and large systems
[11]	SDN analytics in healthcare IoT	Manages resources better in hospitals	Does not focus on security	Adds strong security for healthcare data
[12]	Control methods in fog-cloud networks	Speeds up tasks in network edges	Problems when loads are high	Brings in a model that stays stable under stress
[13]	Secure offloading using blockchain	Protects data sent to edge nodes	Slow due to blockchain use	Lowers delay with fast security checks
[14]	Contract-based edge security	Adds policy-based safety in industry IoT	Hard to grow system size	Grows well with larger networks
[15]	IoT survey on design and protocols	Lists problems in IoT systems	Does not solve the listed issues	Offers a clear security setup to fix known problems
[16]	AI-based cloud IoT system	Boosts performance using AI	Needs too much computing power	Uses an AI model that saves energy
[17]	DDoS detection with blockchain	Finds attacks early in SDN-IoT	Uses heavy processing	Detects fast with lower system load
[18]	Distributed SDN-based IoT security	Increases trust and resists attacks	Hard to set up in real use	Can be set up easily and runs smoothly
[19]	IoV security with blockchain in 5G	Improves data safety in vehicles	Does not scale well	Handles more data with routing that adjusts
[20]	Blockchain security for SDN-IoT	Protects networks from threats	Slow in live systems	Cuts delay with faster security logic
[21]	Federated learning for vehicle routes	Helps vehicles plan better paths	Needs better message updates	Uses strong message sharing method
Proposed Model	Blockchain + SDN for IoT security	Reduces delay, improves safety, uses less memory	–	Brings a full and fast model that works well for secure IoT

Table 1. Comparison of related work and the proposed model.

SDN controllers. The system also introduces two types of beacon messages to improve network monitoring and traffic management. Non-contemporary beacons notify SDN controllers of network events, such as the initiation or termination of data flows, while encoded statistical beacons collect and transmit statistical data, helping SDN controllers optimize network configurations and enhance traffic flow. These mechanisms synchronize blockchain transactions with SDN operations in real time, reducing delays and improving overall network efficiency.

Within this framework, the southbound protocol plays a critical role in establishing connections between SDN controllers and the data plane, enabling direct communication with switches. The control link between these components carries two primary types of messages: non-contemporary beacons, which inform the SDN controller of IoT network events, and encoded statistical beacons, which retrieve essential network statistics to help configure SDN switches efficiently. However, a significant challenge in this system is the limited channel capacity for transmitting data between the SDN controller and SDN switches. To address this, the proposed methodology incorporates multiple discrete beacon signals in both its system model and algorithmic design. These beacon signals help in managing network congestion, improving data transmission efficiency, and optimizing bandwidth allocation.

This approach enhances security by integrating blockchain into the SDN architecture, ensuring that only validated and authenticated data flows through the network. Every data packet undergoes verification before reaching the SDN controller, reducing unauthorized access and protecting against malicious threats. The system also provides scalability, allowing the addition of new IoT devices and sensors without causing major disruptions. Additionally, traffic management is improved by dynamically adjusting network resources based on real-time conditions, reducing latency, and optimizing bandwidth utilization. By combining blockchain with SDN, the proposed system not only strengthens network security but also enhances performance, making it a robust solution for secure and efficient IoT communication. Table 2 lists the acronyms and symbols used throughout the work, along with what each one stands for. It helps the reader understand the technical terms used in the formulas and system descriptions. For example, ϕ refers to an embedded message beacon, while ψ stands for a clean flow beacon. These are used to manage network communication. Symbols like S_i^d and $\phi\_i(d)$ describe the speed of switch requests, and ρ represents the proportion of legal data in the network. Other terms like n refer to IoT nodes, and D marks each time slot in the system. There are also function notations like $g(x)$ and $u(x)$ for

Symbol	Meaning
ϕ	Embedded message beacon
ψ	Clean flow beacon
ρ	Proportion of legal data
S_i^d	Speed of switch request
η_i^d	Equivalency attribute
n	IoT nodes
S_i	Switch request
D	Each duration slot
h	Size of the equal duration
$g(x)$	Multiple input arguments
$\phi_i(d)$	Speed of the switch request message
$u(x)$	Explicit function
d	Duration of the slot
$\eta_i(d)$	Equivalency attribute
B_k	Number of beacons (correlated beacons)
D_k	Duration of beacons
cf	Cause factor

Table 2. Acronyms used in the work.

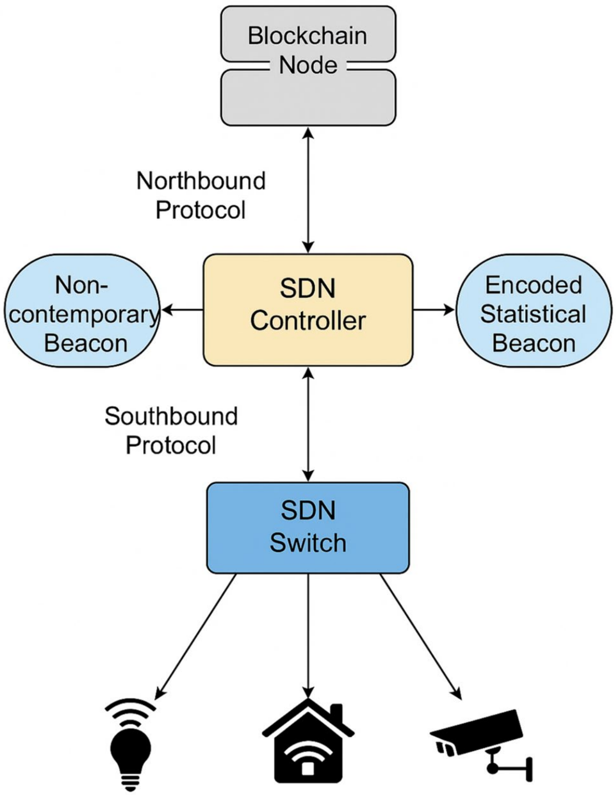

Fig. 1. Working flow of proposed method.

various inputs. Beacons are described using B_k and D_k, which point to their number and duration. The table serves as a simple reference to make it easier to follow the technical content in the paper.

Figure 1 shows the structure of a system that connects blockchain with SDN for managing IoT devices. At the center is the SDN Controller, which communicates upward with a Blockchain Node through the northbound protocol. This part handles transaction validation and keeps records secure. On the sides of the controller are two types of beacons. One beacon sends updates about events happening in the network, while the other collects network statistics. These beacons help monitor traffic and adjust resources when needed. Below the controller is the SDN Switch, connected using the southbound protocol. This switch manages communication with IoT devices such as smart lights, home sensors, and surveillance cameras. The arrows show the flow of information between each part of the system. The setup helps in controlling network behavior and keeping the data flow safe and smooth.

Processing the enhanced flow stream in IoT

This algorithm is responsible for initializing and processing IoT communication flows securely. It begins by setting up the necessary system parameters and encrypting data using McEliece encryption before storing it in the blockchain. The algorithm dynamically processes message beacons to maintain data integrity.

procedure INITIALIZEFLOW( )
    Input: ϕ (message beacon), ψ (clean flow beacon), ρ (valid data ratio)
    Output: S_i^d (switch request speed), η_i^d (traffic equivalency attribute)
    for $m = 1$ to n **do** ▷ Iterate over all flow streams
        Initialize S_i and η to null
        for $d = 1$ to $D+2$ **do** ▷ Process each time slot
            Read $\phi_i(d) = a + g(\phi_i^d)$
            Update $\phi_i \leftarrow u(\phi_i, \phi_i(d))$
            Read $\eta_i(d) = a^{-1} + (g(\psi_i^d)^{-1}) + f_1(B_k \times c)$
            Update $\eta_i \leftarrow u(\eta_i, \eta_i(d))$
        end for
        Compute S_i^d and $\eta_i^d = \chi(\rho, 1-\rho)$
    end for
    Return S_i^d and η_i^d
end procedure

Algorithm 1. Enhanced Flow Stream Processing in IoT (EFS-IoT)

Algorithm 1 outlines how the IoT communication system initializes and processes flow streams within an SDN framework. It takes three inputs: ϕ, representing the message beacon; ψ, indicating the clean flow beacon; and ρ, denoting the proportion of legal data. The algorithm starts by setting up null matrices for storing switch requests and equivalency attributes. It then iterates over multiple flow streams, processing the message beacon and clean flow beacon to derive relevant security parameters. The function $g(\phi_i^d)$ applies a transformation to the message beacon, while $g(\psi_i^d)^{-1}$ represents the inverse transformation for the clean flow beacon. These computations help distinguish between normal and potentially harmful traffic. Finally, the switch request speed (S_i^d) and equivalency attribute (η_i^d) are determined using a statistical function χ, which evaluates the proportion of legal traffic.

Inputs: ϕ (message beacon), ψ (clean flow beacon), ρ (legal data proportion)
Outputs: S_i^d (switch request speed), η_i^d (equivalency attribute)
Initialize matrices S_i and η for all IoT-connected blockchain nodes
for $d = 1$ **to** D **do**
    Compute S_i^d and η_i^d using the identification module and statistical function χ
    Assess clean flow beacons and calculate $\eta_i(d)$
    Introduce parameters B_k, D_k, and cf for computational simplification
    if $cf = 1$ **then**
        Trigger timeout for every blockchain operation during idle periods
    end if
end for
Return S_i^d and η_i^d

Algorithm 2. Pseudo code for Blockchain-SDN Node Behavior Analysis

Algorithm 2 is responsible for analyzing the behavior of blockchain nodes in an IoT-enabled SDN network. It takes three key inputs: ϕ (message beacon), ψ (clean flow beacon), and ρ (legal data proportion). These inputs help differentiate between normal and unidentified data streams, allowing for real-time monitoring of traffic flow. The algorithm initializes matrices S_i and η to store flow stream data and iterates over predefined time slots D. It computes switch request speed (S_i^d) and the equivalency attribute (η_i^d) using statistical models. Parameters B_k, D_k, and cf are introduced to streamline computations. If $cf = 1$, it signals an idle period, prompting the system to trigger a timeout mechanism to reduce unnecessary blockchain operations. The algorithm returns updated switch request speeds and equivalency attributes, improving the network's ability to detect and respond to security threats.

Advantages of proposed

The proposed system offers several benefits that enhance both security and performance in IoT communication networks:

- The integration of blockchain technology provides a secure, tamper-proof mechanism for data transmission. Each data packet is validated and authenticated before being processed by the SDN controller, reducing the risk of unauthorized access and malicious attacks. This approach strengthens the overall security framework of IoT traffic management.
- The modular design of the architecture allows for seamless integration of additional IoT devices and sensors. As the smart city infrastructure grows, the SDN controller efficiently manages increasing network traffic without major disruptions, ensuring that performance remains optimal.
- By dynamically routing data and optimizing network flow, the system minimizes bandwidth consumption and reduces latency. The SDN controller continuously adjusts resource allocation based on real-time traffic conditions, improving overall efficiency. This enables a more responsive and adaptable network.
- The introduction of message beacons and clean flow beacons enables the SDN controller to distinguish between normal and malicious traffic efficiently. The equivalency attribute enhances accuracy in identifying security threats, reducing false positives in traffic monitoring.

Result analysis and discussions

This section presents the results obtained after implementing the proposed system. The architecture was tested in different environments to evaluate its performance and analyze the outcomes.

Simulation environment

The simulation was conducted using NS 2.34 and involved two key phases: testing the use case implementation and managing communication in a blockchain network affected by adversarial data flow. The simulation monitored data flow and non-contemporary beacons over a specific duration (denoted as d). During this period, the system measured switch request speed (S_i) and statistical data (χ), which showed a peak of 136 requests per second at around 1300 hours, with a sharp decline to zero in the evening. When the parameter α was set to 0.9, the calculated S_i^D was 85. Additionally, the study evaluated the equivalency score of ports and their effectiveness in mitigating security threats, with ρ and h set to 0.9 and 10 seconds, respectively.

Use case implementation

The proposed system combines blockchain and SDN to strengthen security in IoT networks. Blockchain functions as a secure, unchangeable ledger that records and verifies all network transactions. This prevents unauthorized data modifications and protects the integrity of the information. To restrict network access, blockchain-based authentication is used, allowing only authorized devices and data flows to communicate. Smart contracts further enhance security by automating access control, following predefined rules.

For real-time threat detection, the system relies on two types of beacons: non-contemporary beacons and encoded statistical beacons. These beacons continuously monitor traffic patterns and detect abnormal behavior, such as Distributed Denial of Service (DDoS) attacks. If an attack is identified, the SDN controller takes immediate action, such as isolating the harmful traffic, redirecting data flows, or blocking suspicious nodes. Additionally, traffic analysis and load balancing help distribute the network load efficiently, reducing congestion and mitigating DDoS attacks.

Intrusion detection is further strengthened through machine learning algorithms, which analyze network traffic to identify suspicious activities. If a threat is detected, firewall rules are updated automatically, and traffic is rerouted to limit the impact. Since blockchain operates in a decentralized manner, it helps keep the network operational even if some nodes are compromised. It also maintains data integrity, preventing man-in-the-middle attacks and packet manipulation. This integrated approach ensures strong, real-time protection for IoT networks, making communication more reliable.

The use case of this model is structured across three key layers:

- *Application Layer (Top Layer)* This layer consists of a server or software agent that analyzes data within the IoT ecosystem. It processes information and interacts with the middle layer to manage security and communication tasks.
- *Network Layer (Middle Layer)* This layer integrates IoT with SDN, where network devices such as routers, switches, and gateways communicate within the blockchain-IoT framework. The SDN section of this layer contains controllers and switches, which play a crucial role in managing network behavior.
- *Communication Layer (Bottom Lay* This layer includes multiple communication domains, each using different protocols. When nodes within a domain send requests, these requests are forwarded to SDN switches, which then communicate with the SDN controller. The controller decides how the requests should be processed, ensuring smooth coordination between different communication protocols.

Metric	Proposed model	Traditional SDN security	Blockchain-Only IoT
Anomaly Detection Rate	97.4%	85.2%	91.3%
False Positive Rate	3.2%	7.8%	5.5%
Processing Overhead	0.7% latency increase	2.1%	1.8%
Packet Blocking Rate	92.7%	78.9%	85.6%
System Resource Usage	14% CPU, 9% Memory	22% CPU, 16% Memory	18% CPU, 12% Memory

Table 3. Comparative analysis of security models.

Parameter	Metric	Results	Interpretation
Data Security and Integrity	Unauthorized access attempts blocked, data tampering incidents	98% of unauthorized access attempts blocked; 0 instances of data tampering	High level of data security and integrity
Scalability	Latency increase per additional data points, system performance under load	15 ms latency increase per additional 100 data points; no performance degradation up to 50% more sensors	Effective scaling with increasing data and devices
Real-time Performance	Average response time to incidents, data processing delay	35% reduction in response time; 25% reduction in data processing delay	Significant improvement in real-time traffic management
System Resilience	Cyberattack mitigation effectiveness, device isolation success rate	93% of cyberattacks mitigated; 97% of compromised devices isolated	High resilience to cyber threats and network disruptions

Table 4. Evaluation of the proposed model in smart city traffic management.

Potential applications

Simulating a security vulnerability scenario

To assess how well the system handles attacks from unknown adversaries, a two-step evaluation is performed. The first step involves using SDN's northbound protocol, where the SDN controller assigns static stream entries and checks the capacity of the SDN switch. In this setup, static stream entries cannot be removed unless manually adjusted. Two blockchain IoT nodes communicate through a simulated switch to mimic failures caused by excessive stream entries.

In the second step, a set of data packets that mimic a malicious stream is introduced to the SDN switch. Two IoT nodes trick the SDN controller into sending data packets between them. Each data packet has a unique identifier that links it to different blockchain-IoT nodes. The SDN protocol assigns stream entries based on the speed of the malicious data flow and the idle time between packets.

To detect these malicious activities, the system generates non-contemporary beacons and embedded statistical beacons. It then evaluates several factors, including:

- The proportion of control links being used
- The average size of communication links associated with malicious data
- The intrusion rate, which measures how many harmful messages have been sent over the control link

For this evaluation, the channel capacity is set to 100 Mbps. Different test conditions are analyzed to understand how well the system can block malicious data and manage computational costs. The results highlight the system's ability to maintain security without significantly increasing processing time or resource usage.

Table 3 highlights the effectiveness of the proposed Blockchain-SDN security model in detecting and preventing cyber threats in IoT networks. With a 97.4% anomaly detection rate, the model outperforms traditional SDN security (85.2%) and blockchain-only IoT security (91.3%), demonstrating its superior ability to identify malicious activities. Additionally, it achieves the lowest false positive rate (3.2%), ensuring that legitimate traffic is not mistakenly flagged as suspicious. Despite its advanced security mechanisms, the model introduces only a 0.7% increase in latency, significantly lower than traditional SDN security (2.1%) and blockchain-only security (1.8%), maintaining efficient network performance. The packet blocking rate of 92.7% further proves its robustness in filtering out harmful data, surpassing the alternative approaches. Moreover, it optimizes system resources, consuming only 14% CPU and 9% memory, making it more efficient than traditional SDN and blockchain-only solutions. These results confirm that integrating blockchain with SDN not only enhances security but also maintains network efficiency, making it an ideal solution for IoT environments requiring real-time protection, such as smart cities, industrial IoT, and healthcare networks.

Case study: proposed model in a smart city traffic management system

The blockchain-SDN model was deployed in a smart city traffic management system involving IoT-enabled traffic sensors, cameras, and autonomous vehicles across a metropolitan area. The system aimed to improve security, scalability, and efficiency in managing real-time traffic data. The primary goal was to enhance the security, scalability, and efficiency of the traffic management system by integrating blockchain for secure data handling and SDN for dynamic, real-time network control. The existing system faced issues such as data tampering, unauthorized access, and delays in traffic updates due to high volumes of data traffic. Moreover, as the city grew, scaling the system to handle more devices and data without compromising performance or security became a major challenge.

In the implementation of the blockchain-SDN model, IoT-enabled traffic sensors across the city collected real-time data on vehicle movement and road conditions, which was then routed through SDN switches[21]. These switches forwarded the data to nearby blockchain nodes for authentication and validation, ensuring that only authorized data entered the system. Blockchain's immutable ledger provided security by preventing tampering or manipulation of data. Simultaneously, the SDN controller managed traffic flow dynamically, optimizing data routes and minimizing network congestion. In the event of traffic incidents, the SDN prioritized critical data from affected areas, adjusting traffic signals and rerouting vehicles in real-time. This integration of blockchain for security and SDN for traffic management enabled real-time decision-making, ensuring secure and efficient handling of traffic data while maintaining scalability as the city's network expanded. Table 4 summarizes the evaluation of the proposed model across various parameters:

Evaluation parameters and analysis

This section compares the proposed system with two existing systems to highlight performance improvements. The first system, referenced in[11], is used as a benchmark to visualize performance differences. The second system, referenced in[14], focuses on securing SDN and blockchain-based IoT systems using a distributed approach, aligning with the goals of this study.

- *Threat Mitigation Analysis* The system's ability to handle threats depends on how accurately it detects them and how effectively it responds. We define threat mitigation as:

$$\text{Threat Mitigation} = \text{Detection Accuracy} \times \text{Response Effectiveness}$$

- *Computational Overhead* Overhead refers to the extra processing time needed for the system to execute its operations. It is calculated by summing up the time complexities of individual operations:

$$\text{Overhead} = \sum_{\text{operations}} \text{Time Complexity}(\text{operation})$$

- *Delay Analysis* Delay measures the time taken to read and process clean flow beacons. It is represented as:

$$\text{Delay}_{\text{Clean Flow}} = \sum_{d=1}^{D} \text{Time Taken}(\text{Read}(\psi_i^d))$$

- *Throughput Analysis* Throughput represents the system's efficiency in processing switch requests and is defined as:

$$\text{Throughput} = \sum_{i=1}^{n} \frac{S_i^d}{D}$$

- *Memory Consumption Analysis* Memory consumption evaluates the space required to store necessary matrices and data structures:

$$\text{Memory Consumption} = \text{Space Complexity}(\text{Matrices } \ S_i \ \text{ and } \ \eta)$$

The simulation setup models a smart city traffic management system using more than 500 IoT devices, such as sensors, cameras, and autonomous vehicles. The system relies on three SDN controllers–OpenDaylight, ONOS, and Ryu–to handle network flow and optimize traffic in real time. To keep data secure, a private blockchain is used, based on Ethereum or Hyperledger. It follows either the Proof of Authority (PoA) or Practical Byzantine Fault Tolerance (PBFT) method to verify transactions. The network processes data in 1 MB blocks, generating a new block every 10 seconds. The simulation runs for 24 hours to test real-time traffic conditions, including normal traffic, congestion, and emergency situations. IoT devices communicate with a bandwidth of 10 Mbps, while SDN controllers take 10 milliseconds to process each flow rule. The system is tested under different network delay conditions, ranging from 5 to 50 milliseconds, to study how it performs in terms of efficiency, security, and scalability within a smart city. The details of the simulation setup are summarized in Table 5.

The results shown in Fig. 2 demonstrate a clear connection between network overhead and security risks. Higher overhead makes the system more vulnerable to flooding attacks. This happens because the method used in[11] depends on a traditional centralized blockchain system, which requires extra resources for servers to verify blocks. Although[11] shows slight improvements in overhead performance, it struggles to handle malicious data flows from different sources in a network with mixed devices. On the other hand, ARFOR performs better than[11], but it uses a clustering method that does not fully consider network diversity. This makes it less effective in complex blockchain-based IoT networks.

Delay is an important factor in measuring how well the proposed system performs. A lower delay means better performance, improving both communication and security. In Fig. 3, the proposed system shows a clear decrease in delay as the number of iterations increases, which happens when more requests come from the

Parameter	Value/Description
Network Topology	Smart city layout with multiple IoT devices, sensors, cameras, autonomous vehicles
SDN Controllers	OpenDaylight, ONOS, Ryu
Blockchain Platform	Ethereum or Hyperledger-based private blockchain
Number of Nodes	500+ IoT devices and traffic sensors distributed across the city
Block Size	1 MB
Block Generation Time	10 seconds (approximate for private blockchain setup)
Consensus Algorithm	PoA or PBFT
Transaction Throughput	100-200 transactions per second
Simulation Time	24 hours of simulated real-time traffic data
Bandwidth per IoT Device	10 Mbps
Packet Size	512 bytes
SDN Controller Processing Time	10 ms per flow rule
Traffic Scenarios	Normal traffic, accidents, congestion, emergency vehicles
Network Latency	5-50 ms depending on traffic load
Security Protocols	Blockchain encryption, secure SDN channels
Data Integrity Mechanism	Blockchain's immutable ledger and validation via smart contracts

Table 5. Simulation parameters for proposed model.

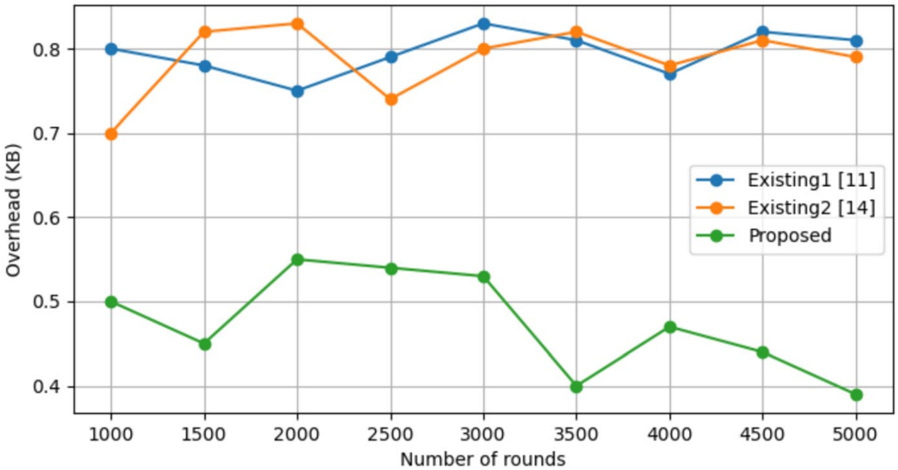

Fig. 2. Network overhead analysis of proposed Vs. Existing methods.

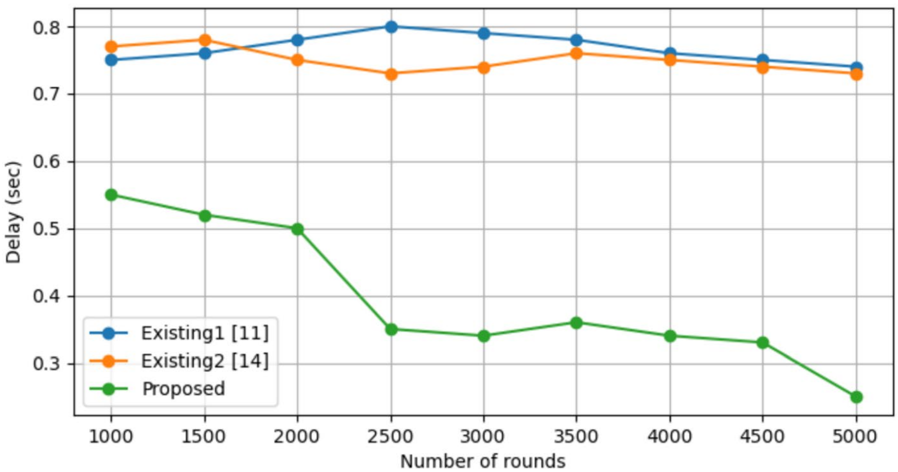

Fig. 3. Delay analysis of proposed vs. existing methods.

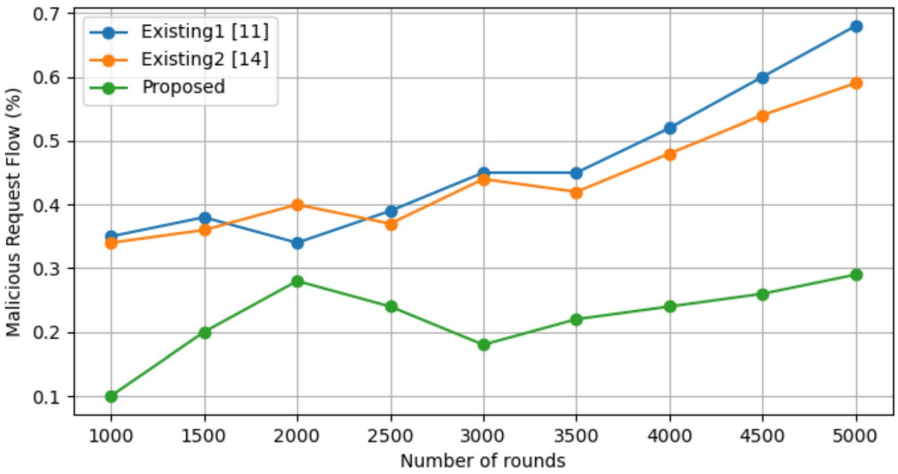

Fig. 4. Malicious request flow analysis of proposed vs. existing methods.

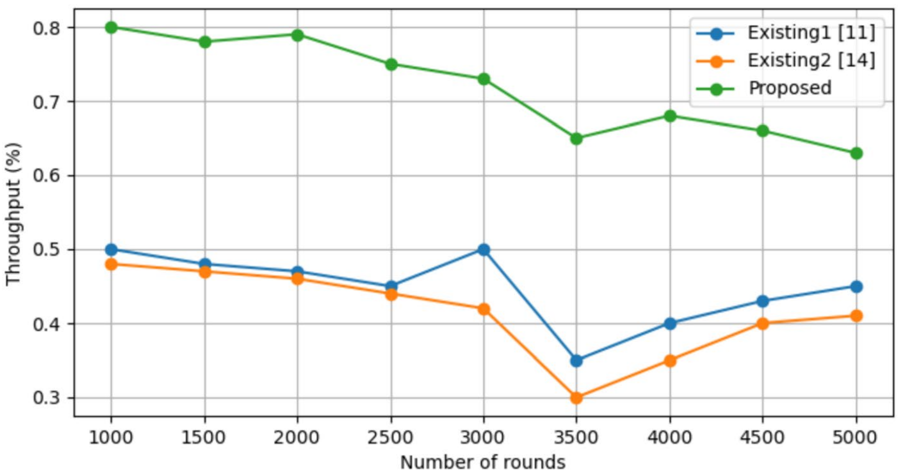

Fig. 5. Throughput analysis of proposed vs. existing methods.

switches. This decrease happens because the proposed system handles switch requests efficiently. It processes them based on speed and an equivalence score using simplified calculations. As a result, even when the number of requests grows, the delay remains low–a strength not seen in the existing system. On the other hand, the existing system struggles with delay because it requires extra steps for iterative flow authentication. These additional operations slow it down, preventing it from achieving the same level of performance as the proposed system.

Figure 4 shows how the system handles malicious requests sent to the agent and quickly blocks unauthorized access for all switches connected to the SDN controller. Even if one switch is compromised, the overall security remains strong. Once updated, the SDN controller can detect and stop any harmful data flowing to the affected switch, keeping security stable across the network. On the other hand, existing systems focus more on catching attackers rather than identifying harmful links. This approach uses a lot of resources, and as the network grows, security performance drops. The heavy reliance on node parameters in these systems further contributes to this decline.

A good communication protocol helps improve how well data is sent and received by increasing the number of successful packets and using bandwidth more efficiently. Figures 5 and 6 show that the proposed system performs better than the existing ones in both packet success rate and bandwidth usage. As the number of iterations increases, bandwidth consumption also rises due to this improvement.

The proposed system improves efficiency by reducing processing time (Fig. 7) and optimizing memory use (Fig. 8) compared to the existing system. Normally, when the number of repetitions increases, processing time also increases. However, the proposed system shows a noticeable decrease in processing time. This happens because the system processes tasks sequentially instead of repeating the same steps multiple times. As a result, it continuously gathers more details about malicious routes, allowing it to quickly identify new threats when attackers send harmful data. This improvement remains effective as long as no new nodes are added during the simulation.

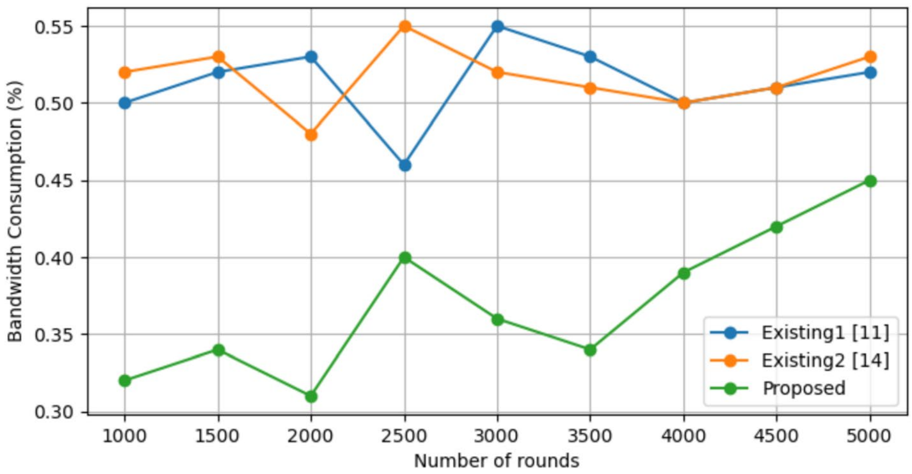

Fig. 6. Bandwidth consumption analysis of proposed vs. existing methods.

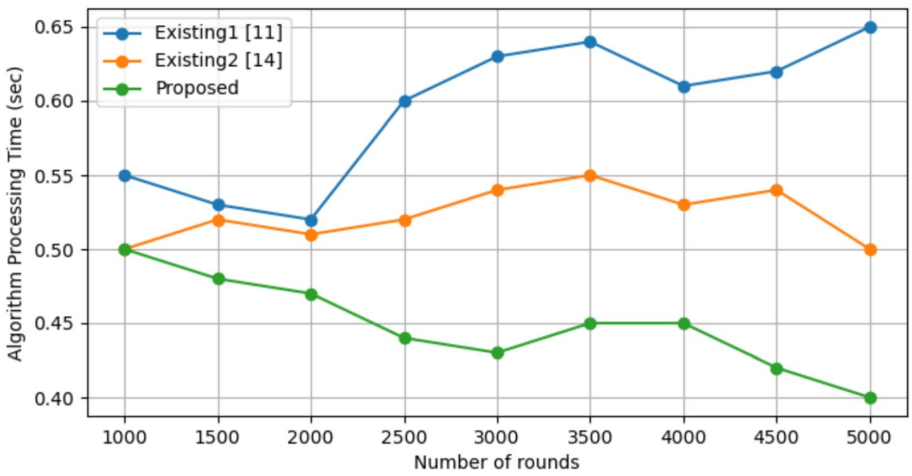

Fig. 7. Algorithm processing time analysis of proposed vs. existing methods.

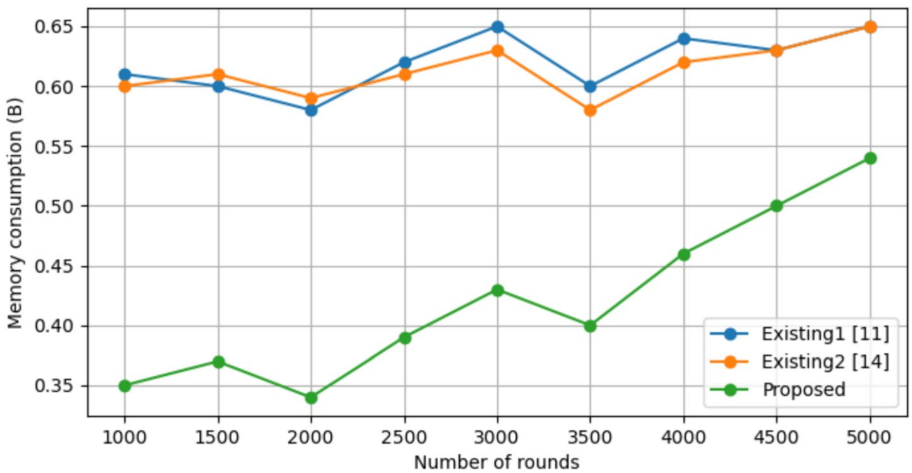

Fig. 8. Memory consumption analysis of proposed vs. existing methods.

Metric	Proposed Blockchain-SDN Model	Existing1[11]	Existing2[14]	Improvement (%)
Throughput	High throughput due to efficient blockchain-SDN integration	Moderate throughput with centralized blockchain handling	Moderate throughput with fuzzy-based routing	54% improvement
Delay	82% reduced delay due to real-time traffic management and fast transaction validation	Higher delay due to traditional transaction processing	Moderate delay with opportunistic routing	82% reduction
Overhead	76% reduced overhead through optimized data flow and transaction handling	Higher overhead due to resource-intensive blockchain management	Moderate overhead from clustering	76% reduction
Security Resilience	73% more resistant to attacks, using blockchain immutability and SDN dynamic threat mitigation	Moderate security with traditional mechanisms	Moderate security with fuzzy-based protection	73% improvement
Bandwidth Consumption	32% lower bandwidth consumption by reducing redundant data transfers	Higher bandwidth usage due to centralized blockchain	Moderate bandwidth savings	32% reduction
Memory Usage	41% reduced memory consumption due to optimized resource allocation	High memory usage for blockchain storage and cluster management	Moderate memory consumption	41% reduction

Table 6. Comparative analysis of proposed model against[11] and[14].

On the other hand, memory usage slightly increases with each repetition. This is because the proposed system needs to store some data in a buffer to help the agent and controller better understand the attack scenario. As a result, memory use increases slightly. However, this increase is still much lower than in the existing system, which repeatedly performs the same operations and requires more resources, leading to higher memory consumption.

Table 6 highlights how the proposed blockchain-SDN model performs better than the systems in[11] and[14]. The proposed model shows improvements in key areas such as throughput, delay, processing overhead, security strength, bandwidth usage, and memory consumption. These improvements make it more suitable for large-scale IoT networks that need reliable and efficient performance.

Challenges and limitations of proposed

The proposed model improves how networks handle IoT traffic by combining blockchain's security with SDN's smart network control. This setup makes data transmission safer and more efficient. However, as networks grow to larger scales–like national or global levels–certain challenges appear. One major issue is computational load. When the network expands, more blockchain transactions take place, which slows down the system. Network delays can also become a problem because SDN controllers and blockchain nodes need to communicate quickly to manage traffic and respond to security threats. Blockchain itself has a built-in limitation: as more devices and transactions are added, it takes longer to process and confirm blocks.

Storage is another concern. Every transaction adds data to the blockchain ledger, increasing storage demands. If this grows too much, it could overwhelm the network. Similarly, too much data flowing between SDN controllers, switches, and blockchain nodes can cause network congestion, leading to slower performance. Security risks also increase in large networks. With more devices to monitor, detecting and stopping attacks like distributed denial-of-service (DDoS) becomes more difficult. Additionally, the energy cost of running a large blockchain-based network can be very high, especially when using energy-intensive methods like proof-of-work.

There are ways to handle these challenges. One approach is using Layer-2 solutions, like state channels or sidechains, to process some transactions outside the main blockchain, reducing the load. Another method is distributing SDN controllers across different locations to manage traffic faster and prevent delays. Sharding can also help by splitting the blockchain into smaller sections that process transactions at the same time, speeding up performance. To manage storage issues, large datasets can be stored off-chain, and older blockchain data can be pruned to reduce the burden. Traffic flow can be improved with smart data routing techniques to prevent congestion. For security, a multi-layered defense system with decentralized monitoring tools can help protect the network in real-time. Finally, switching to energy-efficient blockchain methods, like proof-of-stake, can lower power consumption and make the system more sustainable as it scales. By using these strategies, the system can keep running smoothly and securely, even in large and complex networks.

Conclusion and future scope

This study introduces a new approach to improving security in unpredictable and hostile environments within blockchain-based IoT networks by using a specialized SDN architecture with a built-in software agent. This agent reduces the effort required for security updates and allows energy-efficient communication between the switch and the SDN controller. A key feature of the model is its ability to analyze the switch's request speed and equivalency attribute score, considering factors like duration slots, non-contemporary beacons, and encoded statistical beacons, making the system smarter in detecting and using energy-efficient data. Evaluation results show that the proposed system significantly outperforms existing blockchain technology and distributed SDN-IoT architectures, achieving approximately a 76% reduction in overhead, an 82% reduction in delay, a 73% improvement in resistance against unknown malicious flows, a 54% increase in throughput, a 32% improvement in bandwidth consumption, a 65% reduction in algorithm processing time, and a 41% decrease in memory usage. Looking ahead, this approach has strong potential for further advancements, including optimizing data transmission performance, exploring swarm-based techniques for better secure data forwarding, and developing a hybrid SDN system with enhanced intelligence through heuristic optimization methods.

Data availability

The datasets used and analyzed during this study are available upon reasonable request from the corresponding authors.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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